

Baseline CNN Experiment Report

FaceForensics++ (C23) - MobileNetV2

Objective

The objective of this experiment is to establish a lightweight baseline deepfake detection model against which more advanced multimodal and temporal approaches can later be compared. A frame-based Convolutional Neural Network (CNN) is used to assess whether visual information alone is sufficient for reliable deepfake detection on the FaceForensics++ (C23) dataset.

Dataset and Preprocessing

- Dataset: FaceForensics++ (C23 compression)
- Classes:
 - Real (original videos)
 - Fake (DeepFakes, Face2Face, FaceSwap, FaceShifter, NeuralTextures)

The dataset was split at the video level to avoid data leakage:

Split	Videos
Train	4,200
Validation	900
Test	900

Only videos with successfully extracted frames were included in training and evaluation.

To reduce storage and computational cost while maintaining representativeness:

- 5 frames per video were extracted
- Frames were uniformly sampled across the video duration
- Total frames used:
 - Train: 8,500
 - Validation: 3,555
 - Test: 3,560 (approx.)

This approach aligns with common baseline strategies reported in deepfake detection literature.

Model Architecture

- Backbone: MobileNetV2 (pretrained on ImageNet)

- Modification: Final classifier replaced with a single-output linear layer for binary classification
- Output: One logit per frame (fake vs real)
- Loss Function: Binary Cross-Entropy with Logits (BCEWithLogitsLoss)
- Optimizer: AdamW
- Learning Rate: 1×10^{-4}
- Epochs: 5
- Hardware: CPU

This architecture was intentionally kept simple to serve as a baseline rather than a state-of-the-art detector.

Evaluation Methodology

Since the model operates at the frame level, video-level predictions were obtained by:

$$\hat{y}_{video} = \frac{1}{N} \sum_{i=1}^N \sigma(z_i)$$

Where:

- z_i is the logit output for frame i
- $\sigma(\cdot)$ is the sigmoid function
- $N=5$ frames per video

A video was classified as fake if:

$$\hat{y}_{video} \geq 0.5$$

Results

Test Set Performance (Video-Level)

- Number of test videos: 712
- Accuracy: 73.7%
- ROC-AUC: 0.53

Confusion Matrix

$$\begin{bmatrix} 18 & 146 \\ 41 & 507 \end{bmatrix}$$

Ground Truth	Predicted Real	Predicted Fake
Real	18	146
Fake	41	507

Classification Report

Class	Precision	Recall	F1-Score	Support
Real (0)	0.305	0.110	0.161	164
Fake (1)	0.776	0.925	0.844	548
Accuracy			0.737	712

Analysis and Discussion

The baseline CNN demonstrates strong performance in detecting manipulated videos, achieving a fake-class recall of 92.5%, indicating that most deepfake videos are correctly identified.

However, performance on real videos is notably weaker, with a recall of only 10.9%, revealing a strong bias toward the fake class. This behaviour is consistent with prior research on FaceForensics++, where class imbalance and visually subtle compression artefacts often cause models to over-predict manipulation.

The relatively low ROC-AUC score (≈ 0.53) suggests limited ranking capability across confidence thresholds. This is expected given:

- the absence of temporal modelling
- the use of only five frames per video
- lack of facial alignment or landmark tracking
- and the exclusion of audio and metadata cues

Despite these limitations, the achieved accuracy of 73.7% confirms that even a simple frame-based CNN can capture meaningful visual artefacts associated with deepfake generation.

Conclusion

This experiment establishes a valid and reproducible baseline for deepfake detection using a lightweight CNN. The results highlight both the strengths and limitations of purely visual, frame-based approaches and motivate the need for:

- temporal modelling
- multimodal fusion (audio + video)
- and provenance-aware analysis